Owen McGinty

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Santa Cruz, CA

Education

University of California, Berkeley | *GPA: 3.97* *May 2028*
Bachelor's of Arts | *Mathematics* *Berkeley, CA*
Quantum Computing @ Berkeley, Berkeley Math Torunament, MUSA

Cabrillo College | *GPA: 4.0* *June 2024*
Non-Degree Seeking | *Dual enrollment courses in math, computer science, and humanities*

Aptos High School | *GPA: 4.40* *June 2024*
Awards | *Valedictorian, National Merit Finalist*

Skills

Expertise: Mathematics | Computer Science | Artificial Intelligence | Data Analysis | Simulation | Research
Programming Languages: C | C++ | Java | Python | Lean
Courses (* is grad. course): Number Theory* | Abstract Algebra* | Analysis & Topology* | Algebraic Geometry | Set Theory | Complex Analysis | Linear Algebra | Multivariable Calculus | Discrete Mathematics

Experience

Research Assistant | *Clemson Number Theory REU (Math)* *June 2026 - July 2026 | Berkeley, CA*
Modular forms and elliptic curves

- Mathematics research on the zeros of modular forms with professor Hui Xue
- Proved several interlacing properties for the Serre derivatives of Eisenstein series

Research Assistant | *Prof. Guruswami's TCSLib Project (CS)* *Jan. 2026 - Present | Berkeley, CA*
Formally proving mathematics

- Formally proved essential definitions and lemmas for the analysis of Boolean functions in Lean
- Formally verified several important results on hypercontractivity and threshold functions
- Helped implement new AI-based methods to accelerate the formalization process

Independent Study | *Berkeley Directed Reading & Seminars* *Jan. 2026 - Present | Berkeley, CA*
Mathematics reading courses

- Studied Galois theory and in particular Kummer theory, culminating in a department presentation
- Ongoing study of Lie algebras with a graduate student
- Participating in a research seminar on complex geometry and algebraic topology

Research Assistant | *Prof. Su's HEAT Lab (Nuc. Eng.)* *Nov. 2024 - Present | Berkeley, CA*
Turbulent flow modeling and data analysis

- Simulated fiber optic temperature sensors in extreme sodium-cooled fast reactor conditions via STAR-CCM+, increasing simulation performance by ~2.5x and improving accuracy
- Conducted data analyses of fiber temperature to predict fluid velocity in Python
- Pursued novel research on near-wall behavior of molten salt and its impact on accurate velocimetry

Mathematics Instructor | *Descartes Learning Club* *Jan. 2026 - Present | Berkeley, CA*
Accelerated mathematics instruction

- Teaching advanced mathematics from the Art of Problem Solving curriculum to K-6 students
- Producing individualized lesson plans and designing math puzzles